Vasu Singhal

Embedded Systems Engineer – R&D & System Integration
Computer Vision · RFID · Access Control · ANPR · GSM OTP · Gate Automation

Open to Work – Immediate Joiner

Noida, UP | vasusinghal2018@gmail.com | LinkedIn | Portfolio | GitHub

SKILLS

Computer Vision & AI
OpenCV · InsightFace · ArcFace · dlib · MTCNN · Face Embedding · ANPR / LPR · Tesseract OCR · Edge AI

Embedded & IoT
Arduino · ESP8266/32 · NodeMCU · Raspberry Pi · MFRC522 · Mifare 1K · SIM800L/900A · SPI/I2C/UART · RS-232/485

Software
Python · C / C++ · SQLite/MySQL · Flask API · RTSP · Multithreading · Tkinter/PyQt · AT Commands

Access Control Hardware
Touch Kiosk · Flap Barrier · Boom Barrier · EM Lock · Relay Circuit · Anti-passback

Operations
MS Excel · MIS Reports · Inventory Mgmt · GST Tracking

EDUCATION

B.Tech – ECE
ABES Engg. College (AKTU)
76.6% · 2023

Diploma – Digital Electronics
BTE Delhi
77.89% · 2020

10th – CBSE
6.8 CGPA · 2017

LINKS

- vasusinghal45.github.io
- github.com/vasusinghal45
- linkedin.com/in/vasu-singhal
- vasusinghal2018@gmail.com

SUMMARY

Embedded Systems Engineer with **2+ years** at Socteamup Semiconductor, Noida — building **AI-powered access control & security systems** end-to-end. Shipped **14 production systems** across libraries, pools & society gates using InsightFace/OpenCV, RFID, ANPR, GSM OTP, and barrier hardware. Expert at hardware-software co-design — from firmware to real-time dashboards. **Available for immediate joining.**

EXPERIENCE

Socteamup Semiconductor Pvt. Ltd.

Embedded Systems Engineer – R&D & System Integration

April 2024 – Present | Noida, Uttar Pradesh

Face Recognition – Attendance & Library Kiosk (v1 → InsightFace Upgrade)
Built real-time pipeline (OpenCV+dlib HOG, 128-d embedding) → migrated to **InsightFace/ArcFace+MTCNN** achieving **sub-100ms inference**. Deployed industrial touch-screen kiosk at library — zero-touch attendance, SQLite records, member fine display. Eliminated 100% manual punch-in/out.

InsightFace · ArcFace · OpenCV · dlib · Python · SQLite · Touch Kiosk

Multi-modal Kiosk + Flap/Boom Barrier + Live Dashboard
Extended kiosk with **RFID MFRC522 (SPI, Mifare 1K)** + barcode (QR/EAN-13). Built RTSP/Flask admin dashboard (member DB, fines, attendance export). Integrated **flap barrier via RS-485 relay + anti-passback logic** — deployed at library & swimming pool.

MFRC522 · SPI · Flask API · RTSP · MySQL · Flap Barrier · RS-485

Custom RFID R/W – Encrypted Sector/Block Protocol
MFRC522 at Mifare sector/block level with **Key A/B auth** — binary encoding (member ID, access level, checksum) across sectors. Cards are **anti-clone & offline-verifiable**; write-protected via access bit configuration.

MFRC522 · Key A/B Auth · Anti-clone · Binary Encoding

ANPR / LPR – AI Vehicle Number Plate Recognition & Gate Control
OpenCV pipeline (Canny edge + adaptive threshold + morphological ops → plate localization) + **Tesseract/EasyOCR**. Whitelist DB triggers **boom barrier open or security alert** for vehicles.

OpenCV · Tesseract OCR · EasyOCR · ANPR/LPR · Boom Barrier

GSM OTP Gate + Smart Visitor Kiosk – Multi-factor Physical Access (MFA)
SIM800L AT commands → auto voice call + SMS OTP to resident → TOTP validation → relay to boom barrier (<30s flow). Next-gen visitor kiosk: **face recognition + flat lookup + OTP → EM lock + flap + boom** via unified middleware. Full visitor audit trail.

SIM800L · TOTP/OTP · MFA · InsightFace · EM Lock · Boom Barrier · Audit Trail

Inventory Management, MIS Reporting & Expense Tracking
Excel-based inventory system (inward/outward logs, reorder alerts, vendor history). MIS reports with PivotTables, VLOOKUP, dynamic charts for budget vs. actual tracking. GST expense tracker with monthly reconciliation reports for financial audit readiness.

MS Excel · PivotTable · MIS Reports · GST Tracking · Inventory Mgmt

CERTIFICATIONS

Front End Development – **Great Learning**
C Programming – **TCS iON**
Internet of Things – **Emertxe**
Embedded System & IoT – **ABES**
CCNAv7: Intro to Networks – **Cisco**
TCS iON Career Edge – **Young Professional**

HIGHLIGHTS

- Delivered **5 production systems** at real client sites within 12 months
- Robotics (O-Level) – Times Institute of Mgmt. & Technical Studies
- Embedded System Design MOOC – Texas Instruments
- Academic: Temp-Controlled Fan | Soldier Health Tracker | IoT Home Automation (Arduino, NodeMCU, Blynk)